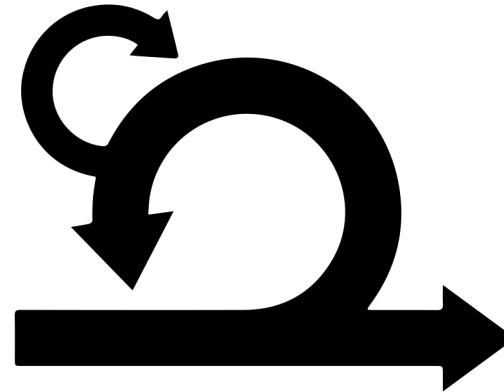




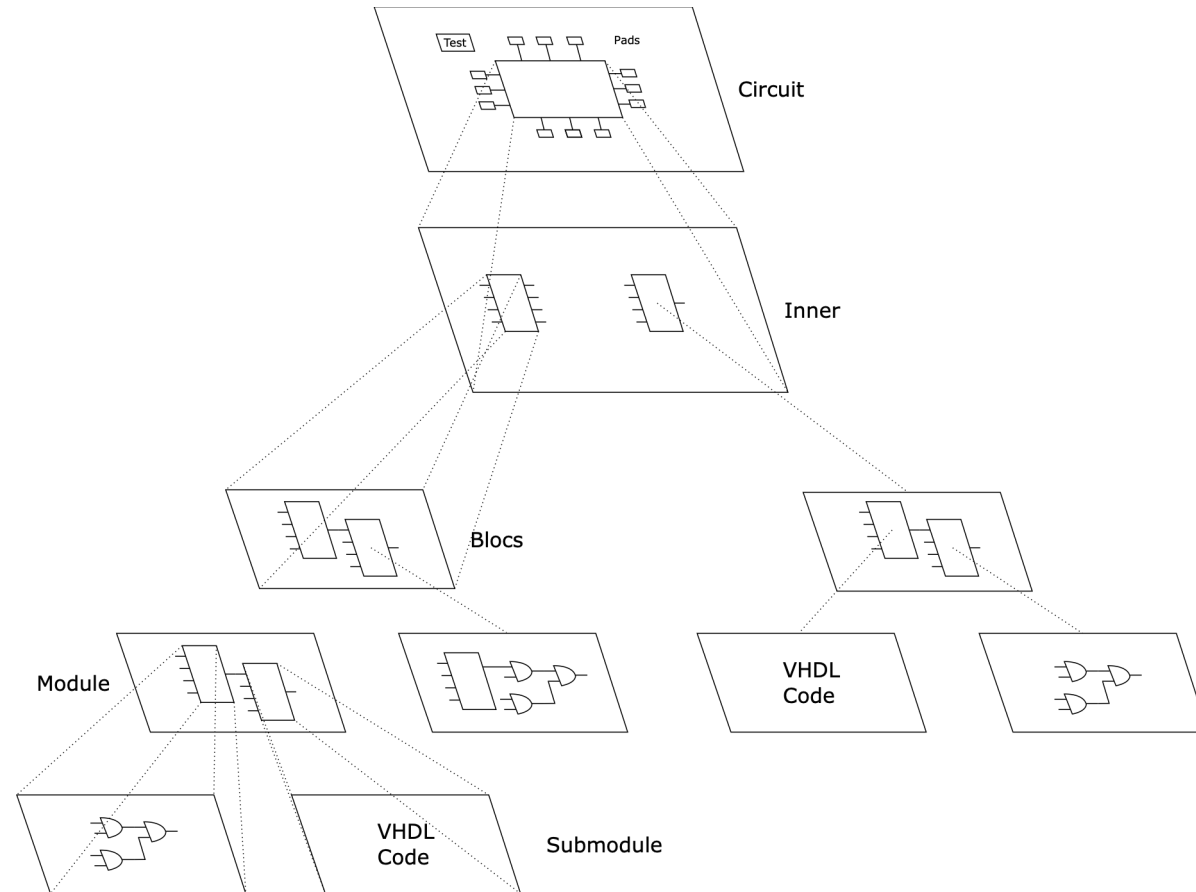
System Design Technical Management

Systems Engineering program



Silvan Zahno silvan.zahno@hevs.ch

What for and why?



Technical Management Methods



- PRINCE2
- SixSigma
- Lean
- Critical chain project management (CCPM)
- Waterfall / Diagram-V / 6q
- Extreme programming (XP)
- Scrum / Kanban

Technical Management Methods



- PRINCE2
 - SixSigma
 - Lean
 - Critical chain project management (CCPM)
- Industry

- Waterfall / Diagram-V / 6q
- Development

- Extreme programming (XP)
 - Scrum / Kanban
- Software

Technical Management Methods

PRINCE2

Organized approach to achieving project goals within constraints.

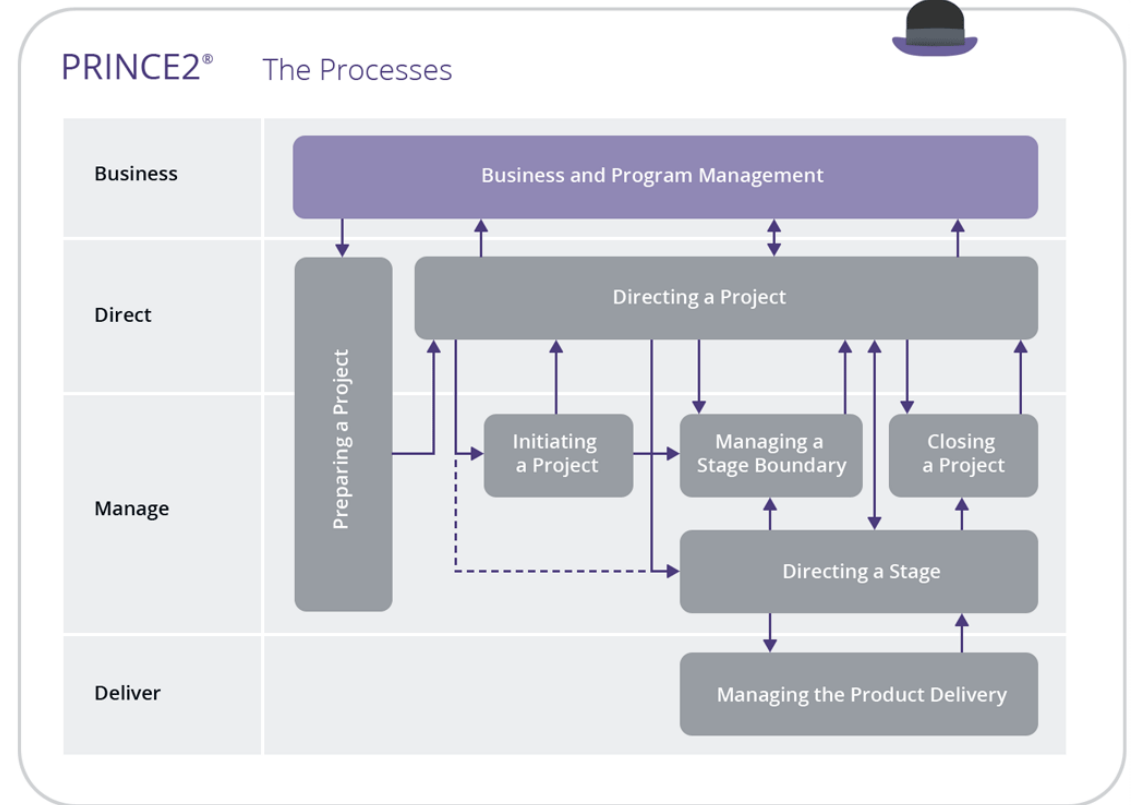
What it is:

Used for quality management to eliminate project defects.

Who should use it:

Large teams that tackle complex projects

Example: BBC, UK Government



<https://www.microtool.de/en/knowledge-base/how-does-prince2-work/>

Technical Management Methods

Six Sigma

Six Sigma is a method for improving process quality by reducing variability and eliminating defects.

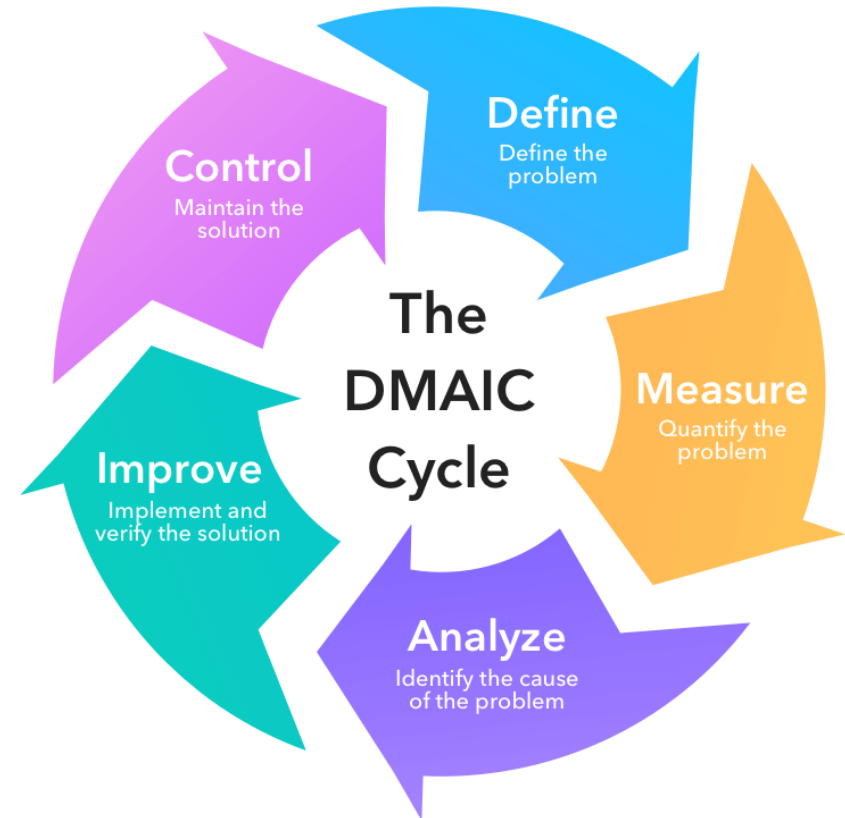
What it is:

Used for quality management to eliminate project defects.

Who should use it:

Large organizations looking to reduce errors

Example: Motorola, General Electric



<https://startinfinity.com/project-management-methodologies/six-sigma>

Technical Management Methods

Lean (Production)



Lean is a methodology for optimizing process efficiency and reducing waste by focusing on adding value from the customer's perspective and continuously improving the process flow.

What it is:

Aims to cut waste and create a simple framework for project needs

Who should use it:

Teams struggling with efficiency issues

Example: Toyota, Amazon



<https://www.sketchbubble.com/en/presentation-lean-project-management.html>

Technical Management Methods

Extreme Programming (XP)



Extreme Programming (XP) is an agile software development approach that focuses on frequent releases, continuous testing, and customer feedback to deliver high-quality software products.

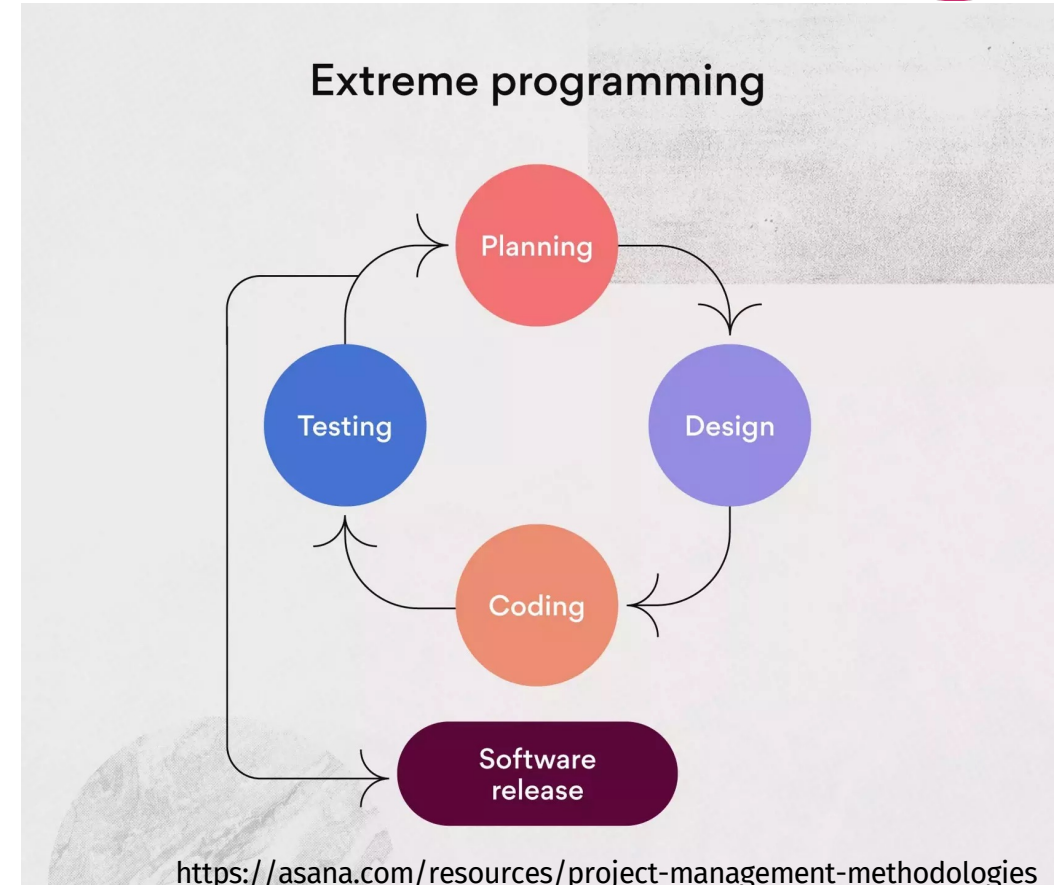
What it is:

Works by creating short development cycles with many releases

Who should use it:

Small top mid-size teams that have tight turnaround times

Example: Spotify, Netflix



Technical Management Methods

Scrum

Is an agile framework that organizes work into fixed time intervals (sprints) with defined roles and ceremonies to improve team coordination and deliver incremental value.

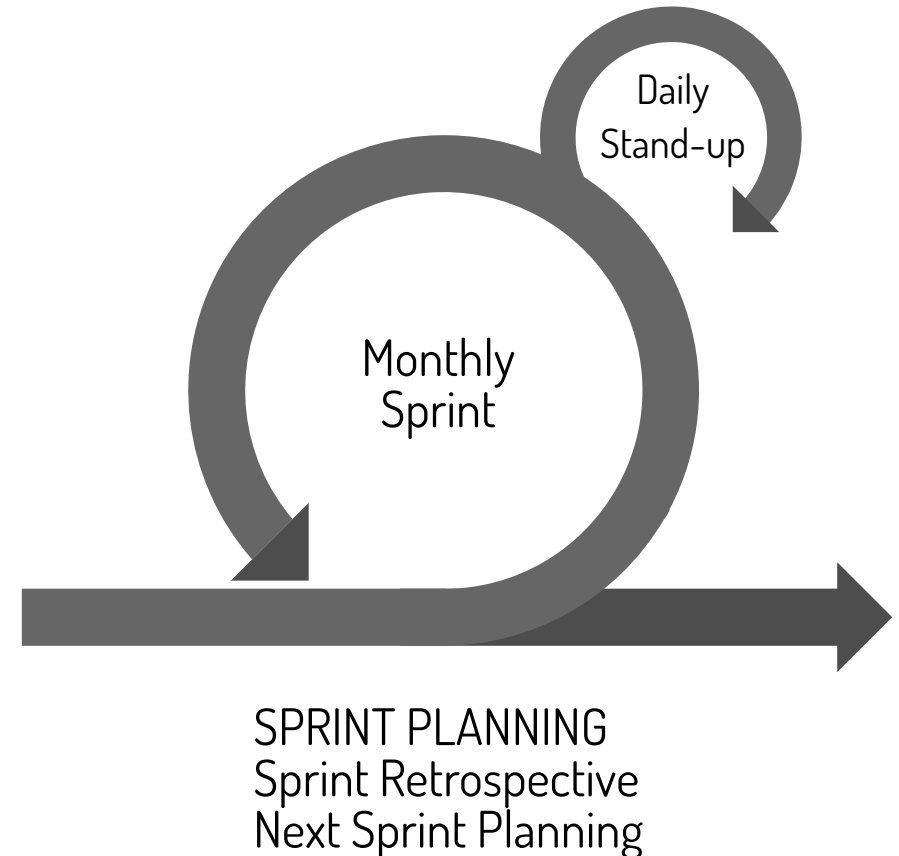
What it is:

Breaks projects into sprints with daily standups, backlog refinement, and sprint reviews. Each sprint has a clear goal; progress tracked on burndown charts. Roles: Product Owner (prioritizes work), Scrum Master (removes blockers), team (executes).

Who should use it:

Teams of 5-10 that need visibility, flexibility, and regular stakeholder feedback. Works well when requirements evolve.

Example: Google, Microsoft, AirBnB





Waterfall / Diagram-V / 6q

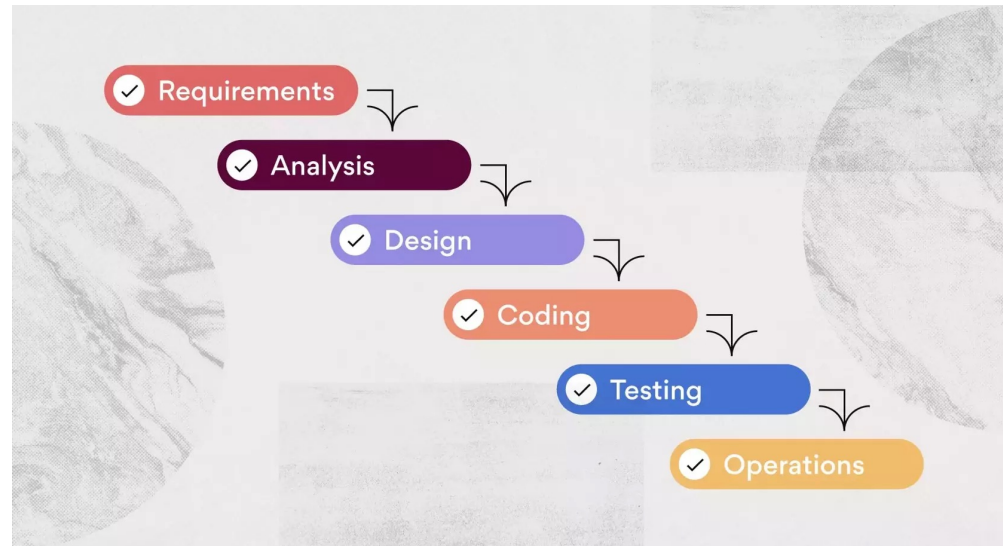
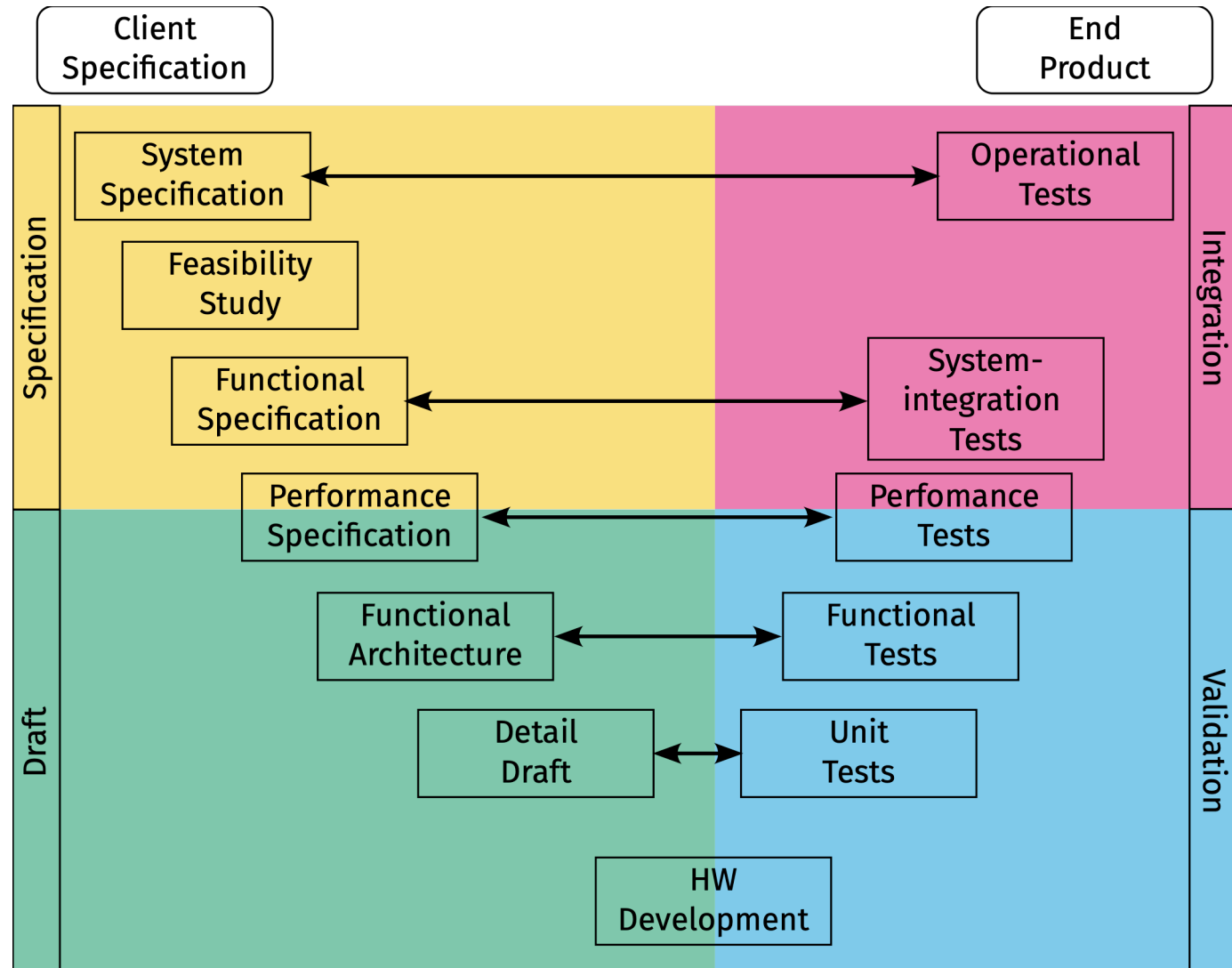
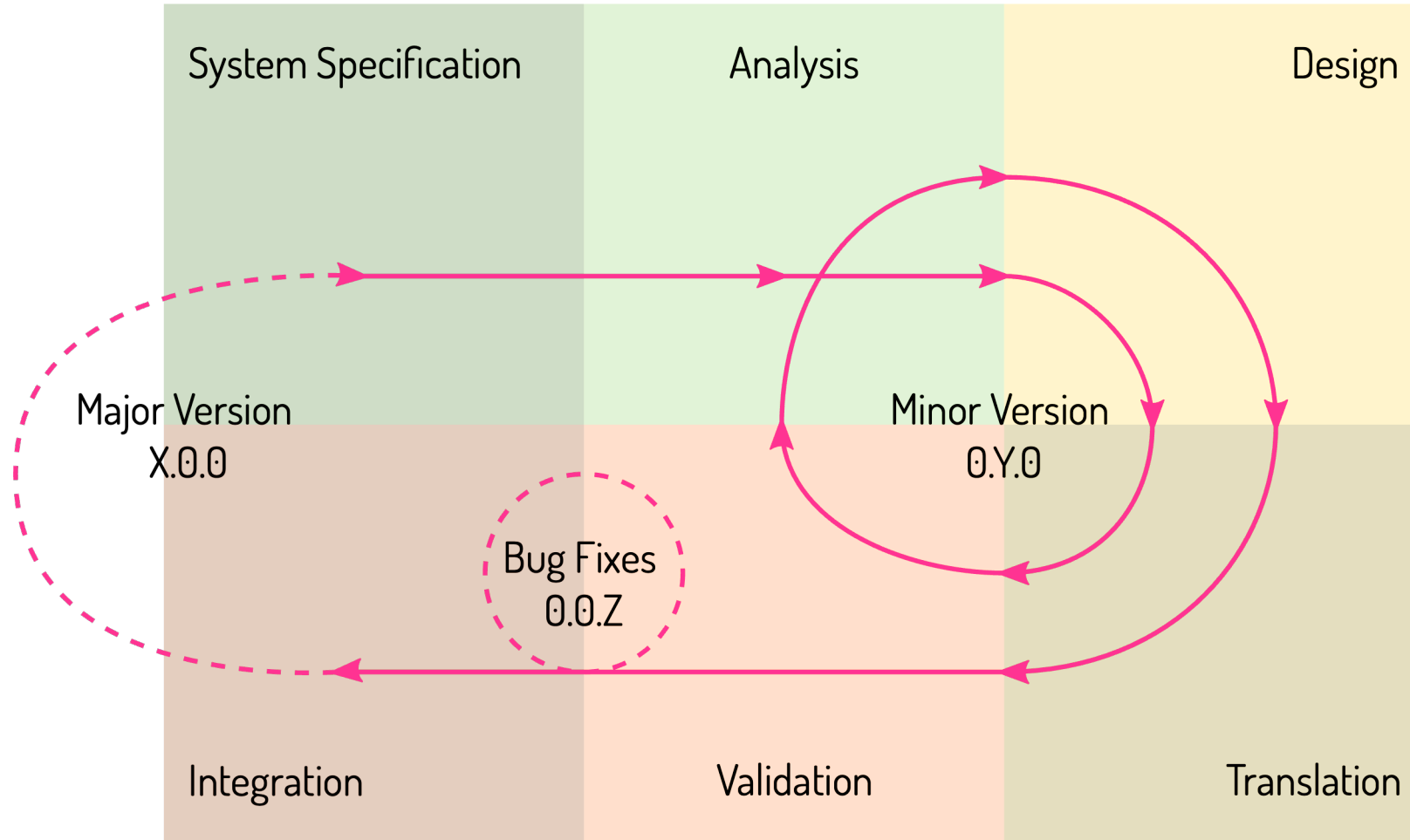


Diagram-V



6q



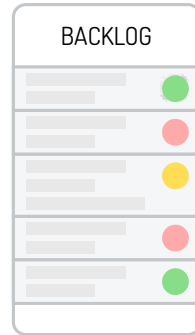


Agile
Scrum / Kanban

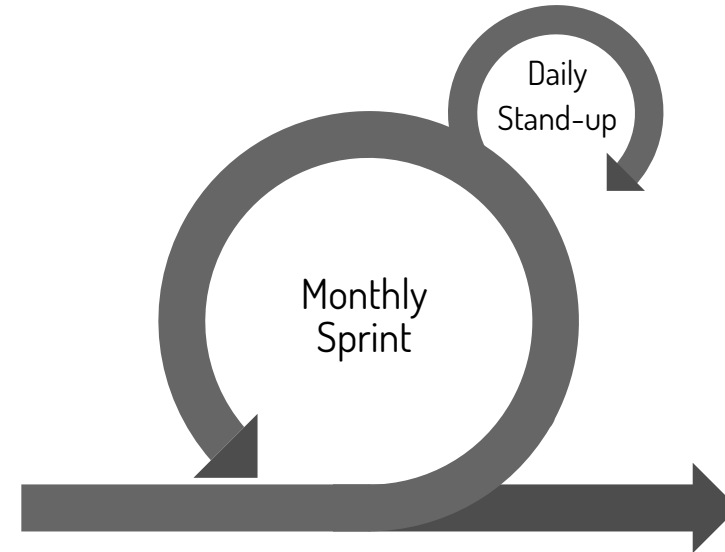
Scrum



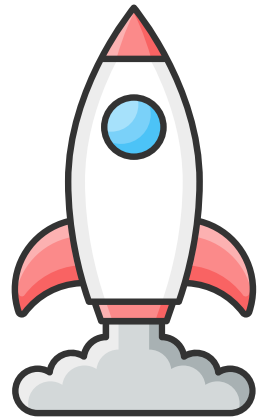
IDEA / VISION
Business Model
Milestones
Releases



PRODUCT BACKLOG
Project specific
prioritized workpackages



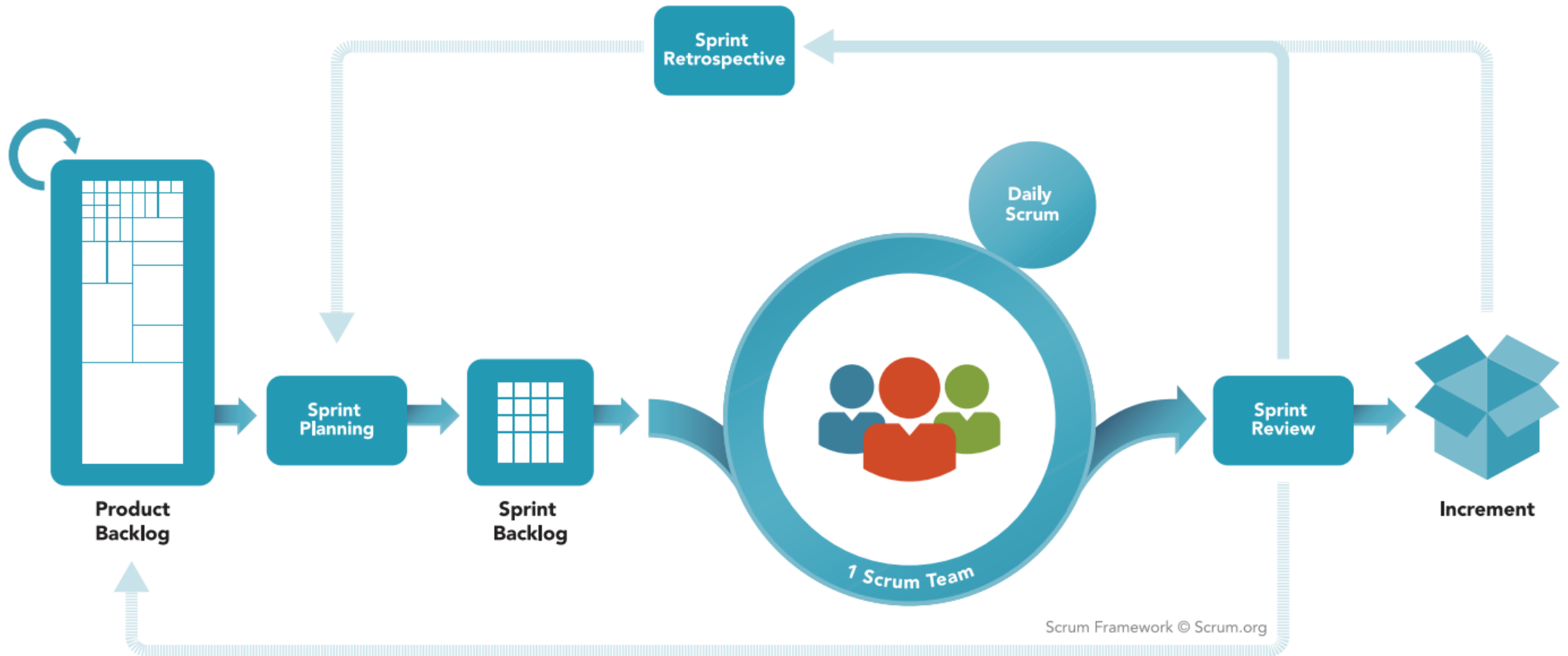
SPRINT PLANNING
Sprint Retrospective
Next Sprint Planning



FINISHED PRODUCT

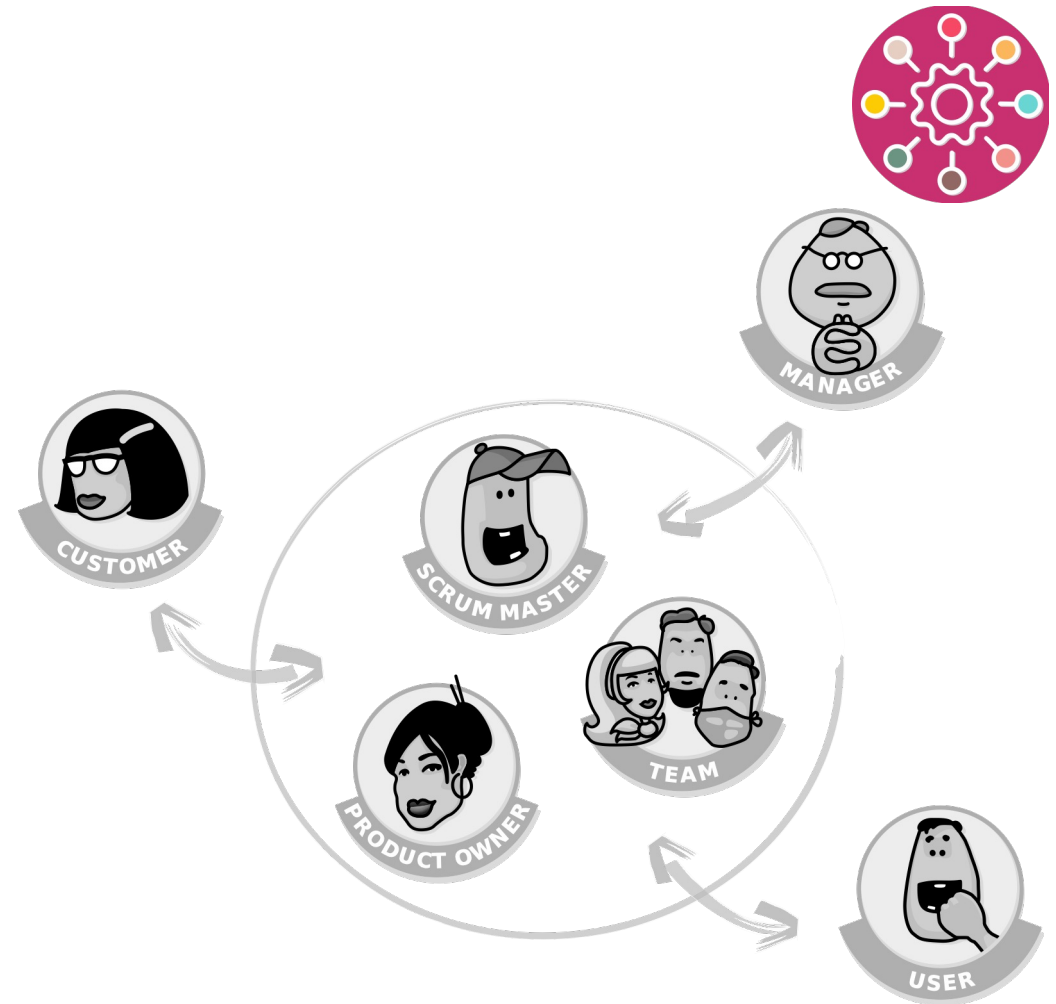


Scrum



Practical Scrum

- The Roles in the Scrum Team
 - Product Owner
 - Development Team
 - Scrum Master
- The Scrum artefacts
 - User Story
 - Product Backlog
 - Sprint Backlog
 - Increment
 - Definition of Done, Definition of Ready

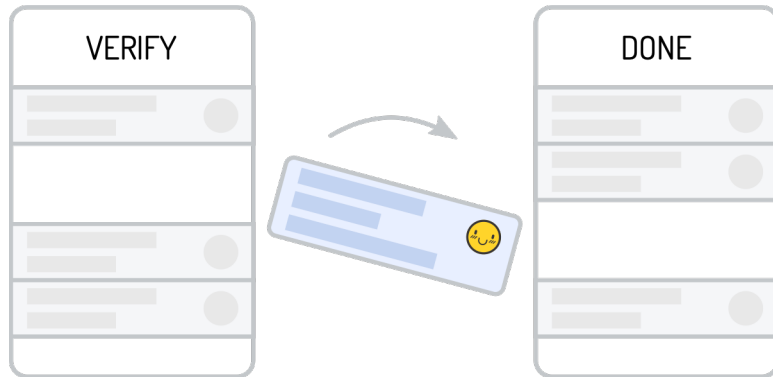


B. Gloger and K. Schwaber, *Scrum: Produkte zuverlässig und schnell entwickeln*, 4., Überarb. Aufl. München: Hanser, 2013.

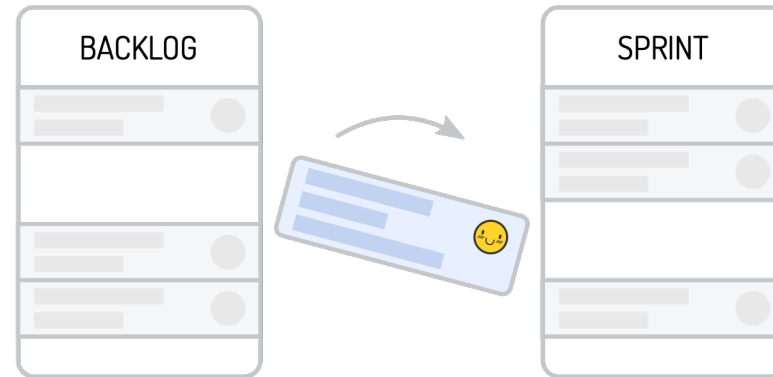
Scrum Events



Sprint Retrospective



Sprint Planning



User Story



Explicit Title

- e.g. for a Taxi ride app: "Search for an available Taxi"

Narrative phrase: "As [persona], I want [feature], so that [function]"

- e.g. "As a customer, I want to see all nearby taxis, so that I can select one"

Acceptance Criteria

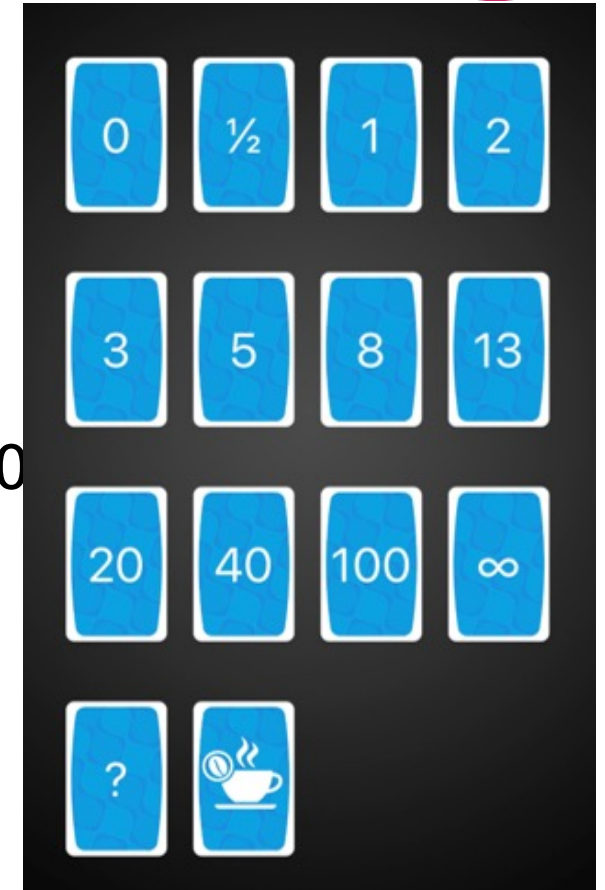
- e.g. "Only display available taxis"
- e.g. "Display them on a list from the closest to the furthest away"
- e.g. "Show a picture of the car"

Task Estimation



Scrum Time App

- Within the team 1 points needs to be defined and refined.
- Sequences
 - Planning Poker: 0, $\frac{1}{2}$, 1, 2, 3, 5, 8, 13, 20, ..., 100
 - Fibonacci: 0, 1, 2, 3, 5, 8, 13, 21, 144
 - Natural: 1, 2, 3, 4, 5, 6, ..., 16
 - T-Shirt: XS, S, M, L, XL

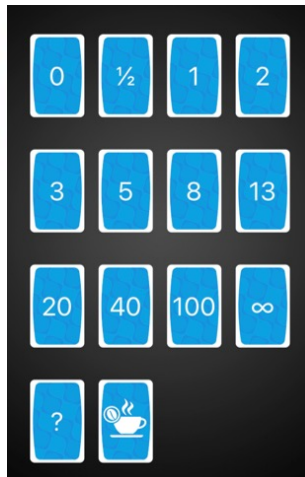


Task Estimation

Example 1

Search a Lego in a Pile and
stack is corresponds to:

1/2 Point

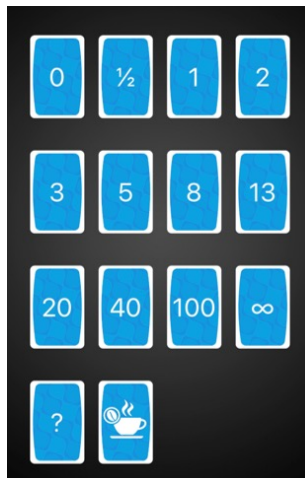


Task Estimation

Example 2

Create a small Lego robot

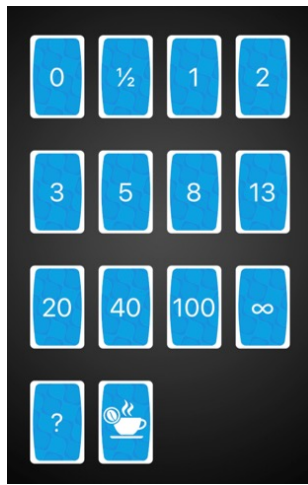
5 Points



Task Estimation

Estimation 1

Create a big Lego robot
X Point(s)



Kanban



OctoArcade Invaders

Planning Sprint Board Alpha Roadmap My work Features Priority By person Status Board By status By Sprint Done

Filter by keyword or by field

Not Started 19 Estimate: 37

- planning-tracking-demo #810
Beta go-no-go meeting
- planning-tracking-demo #800
Save score across levels
- planning-tracking-demo #784
Interviews with media outlets
epic
- Draft
Enable for teams
- planning-tracking-demo #1161
tweak difficulty
- planning-tracking-demo #1167
Update README.md
- Draft
Prevent the Konami code from bringing down all of GitHub

+ Add item

Planning 19 Estimate: 109

- planning-tracking-demo #823
Updates and bug fixes to engine from Beta
bug demo
- planning-tracking-demo #824
Beta signup page
need help
- planning-tracking-demo #806
[Tracking] Upsell / Growth experience
backlog feature
- planning-tracking-demo #818
Account subscription design
- planning-tracking-demo #828
Acquire domain for launch
- planning-tracking-demo #832
Final creative shots from game
- planning-tracking-demo #829

+ Add item

Building 8 Estimate: 40

- planning-tracking-demo #1160
Update documentation
- planning-tracking-demo #814
Updates to collision logic
enhancement
- planning-tracking-demo #816
Free and paid levels
need help
- planning-tracking-demo #831
Documentation and Support
need help
- planning-tracking-demo #821
Updates to alien, beam, bomb and cannon sprites
#370
- planning-tracking-demo #802
Updates to velocity of the ship and alien movements

+ Add item

Review 5 Estimate: 17

- planning-tracking-demo #822
Hero site - Development
#12 #1160 in-review task urgent
- planning-tracking-demo #808
General bug fixes from Alpha feedback
#992
- planning-tracking-demo #1151
Design new launch screen
#374 web
- planning-tracking-demo #793
Polished alien, beam, and cannon sprite files
- planning-tracking-demo #1101
[Tracking] Integrate payments system
backlog feature

+ Add item

<https://github.com/features/issues>

DoR – Definition of Ready (Example)



- Story written as a "User Story" (i.e. “As ... I want ... so that ...”)
- Acceptance criteria must exist and be understood by team
- User Story has been estimated by the team (in points)
- Where appropriate: Sketches, diagrams, annexes, reference documents and other resources exist, on a shared access platform
- The User Story is validated by the Product Owner
- The User Story is validated by the Dev Team

DoD – Definition of Done (Example)

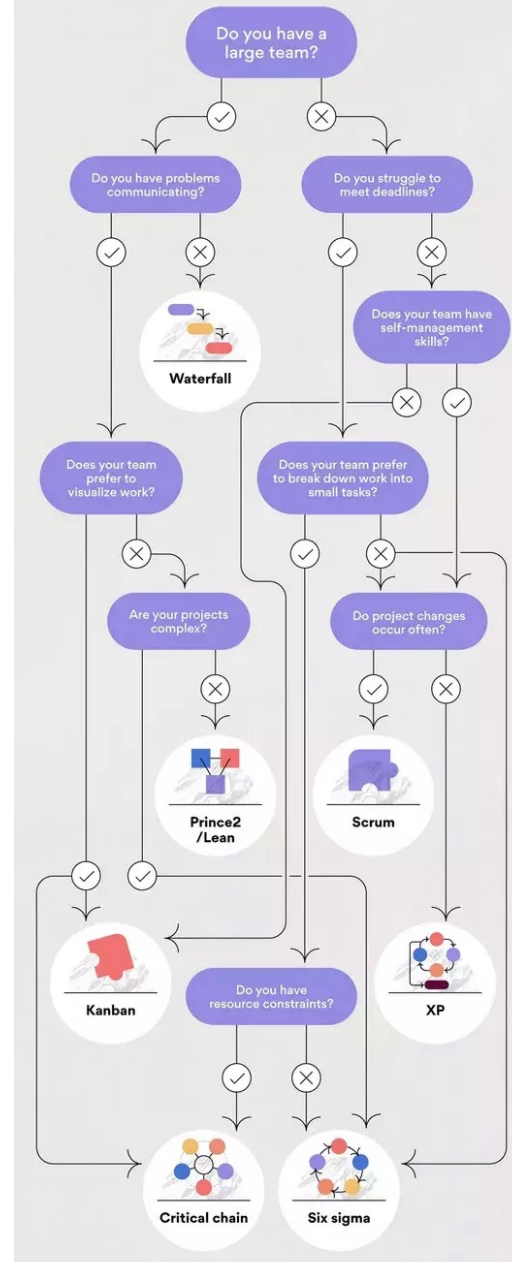


- If it is a code development User Story: The code builds and is pushed to the git repo
- If it is a code development User Story: The code has been reviewed and merged in a "master branch" on git
- If it is a code development User Story and if applicable: Some automated tests (checking for the User Story's Acceptance Criteria) are running on git
- If it is a code development User Story and if applicable : A documentation is available either as a ReadMe file in the repo or in the general documentation
- If the user story is not code related, All generated document are archived
- All the User Story's Acceptance Criteria are met and verified by the Team (e.g. in the Sprint Review)
- The User Story is accepted by the Product Owner



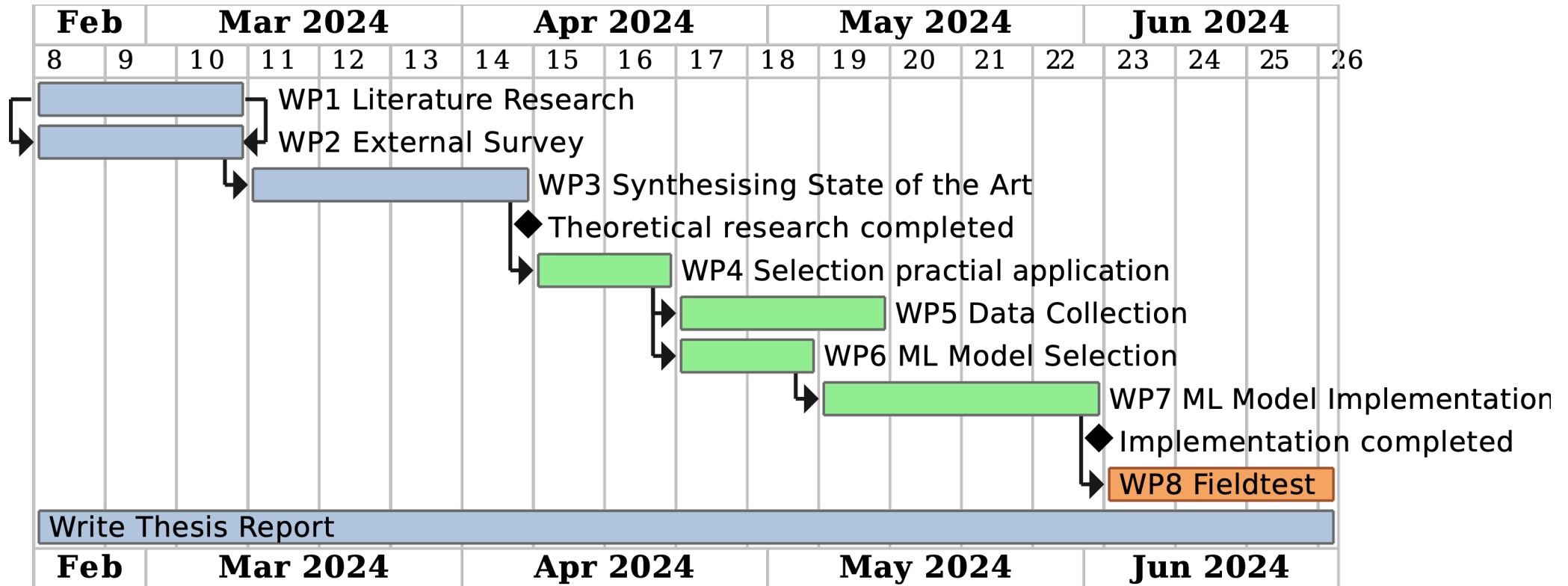
Outro

What is the right method ?

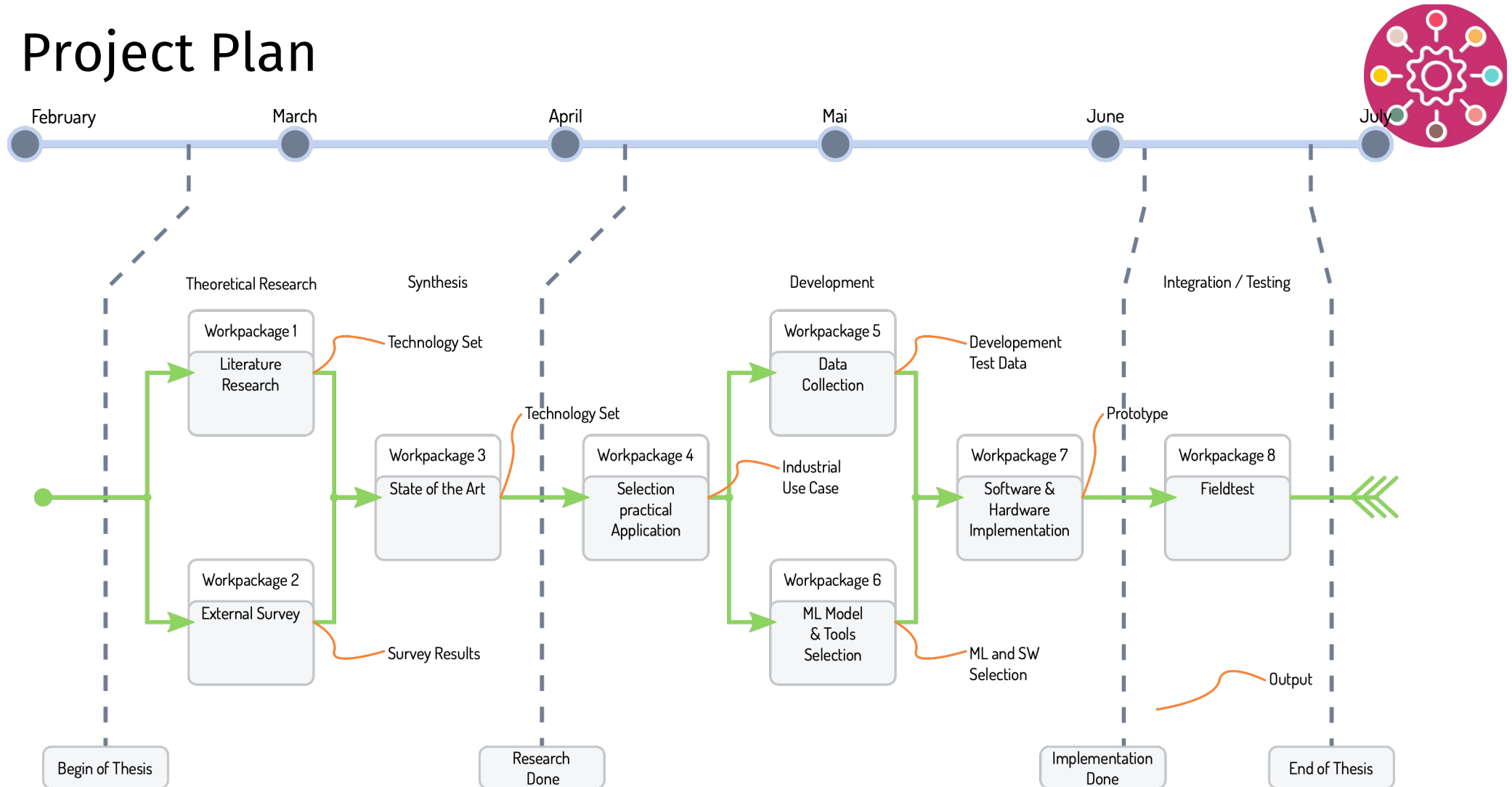


<https://asana.com/resources/project-management-methodologies>

Gantt-Diagram



Project Plan

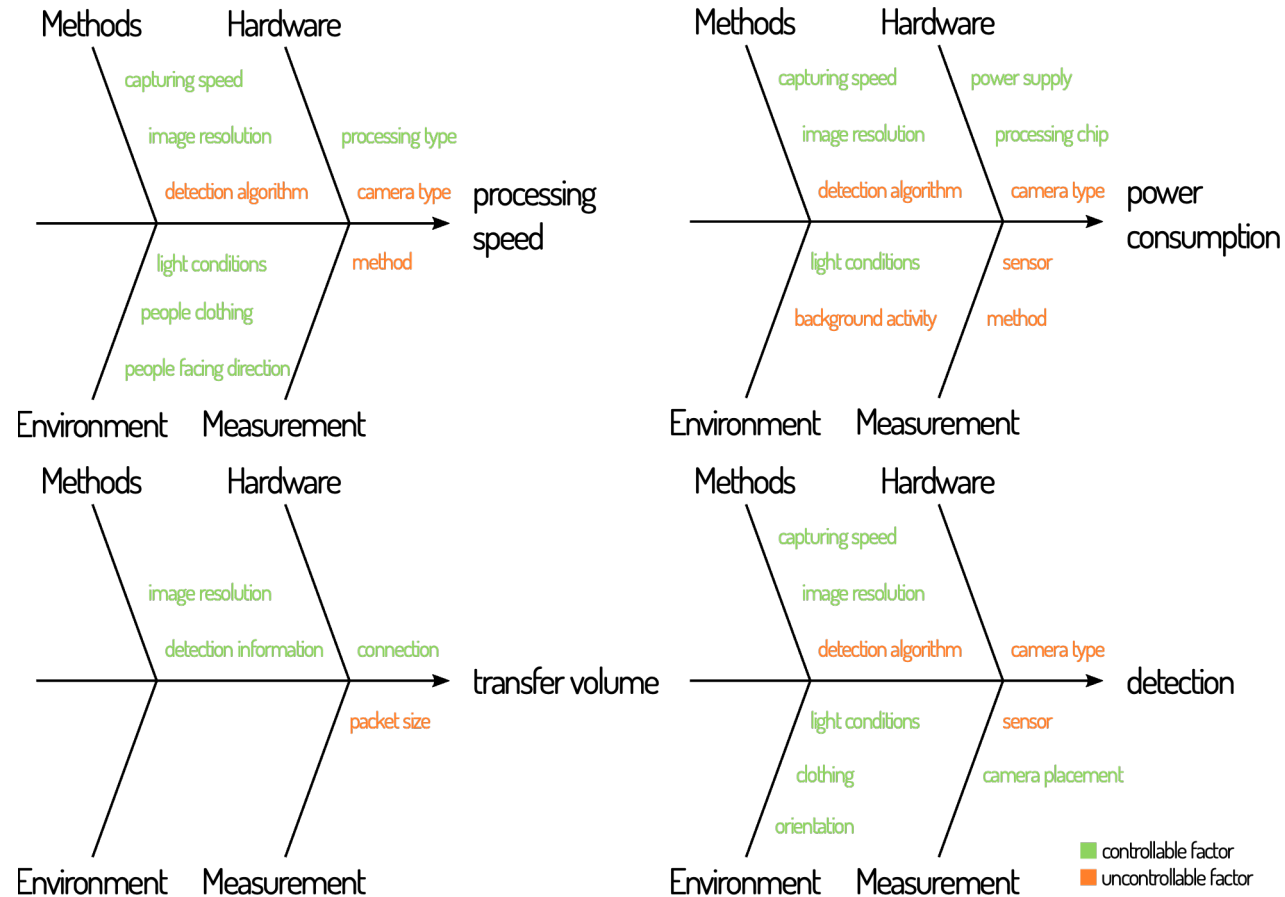


SWOT-Analysis

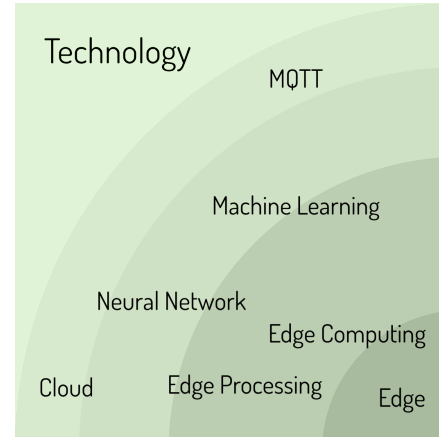


	Helpful for your objective	Harmful for your objective
Internal within technology	Strengths • On-site processing, reducing bandwidth and latency • Longer hardware lifecycle • Realtime capabilities • Adaptive HW for future expansion • Reliable HW for security applications	Weaknesses • Many ML nets are implemented and tested on other hardware first • Different programming and HW knowhow required • Custom hardware necessary depending on the application
External outside technology	Opportunities • Other types of algorithms can be implemented • Development of a new ecosystem • Proven technology also used in datacenters, satellites and medical	Threats • CPU and GPU technologies are getting faster and cheaper • User resistance to change to a new technology • Industrial sector not used to • Smaller hardware choice available

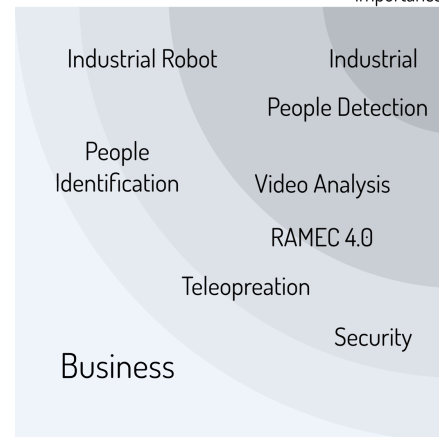
Cause and Effect Diagram



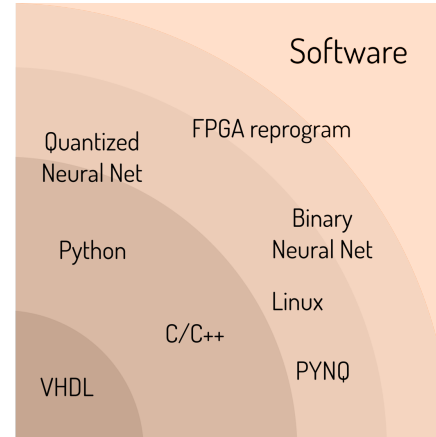
Keyword Radar



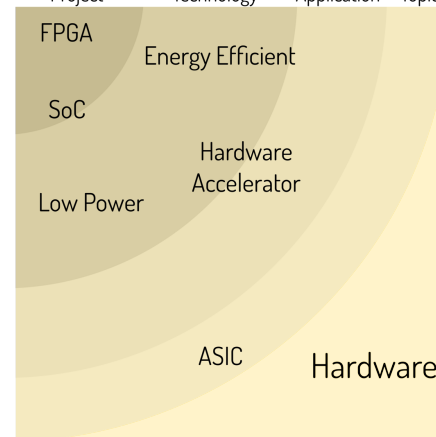
Importance



SyD



Project Technology Application Topic

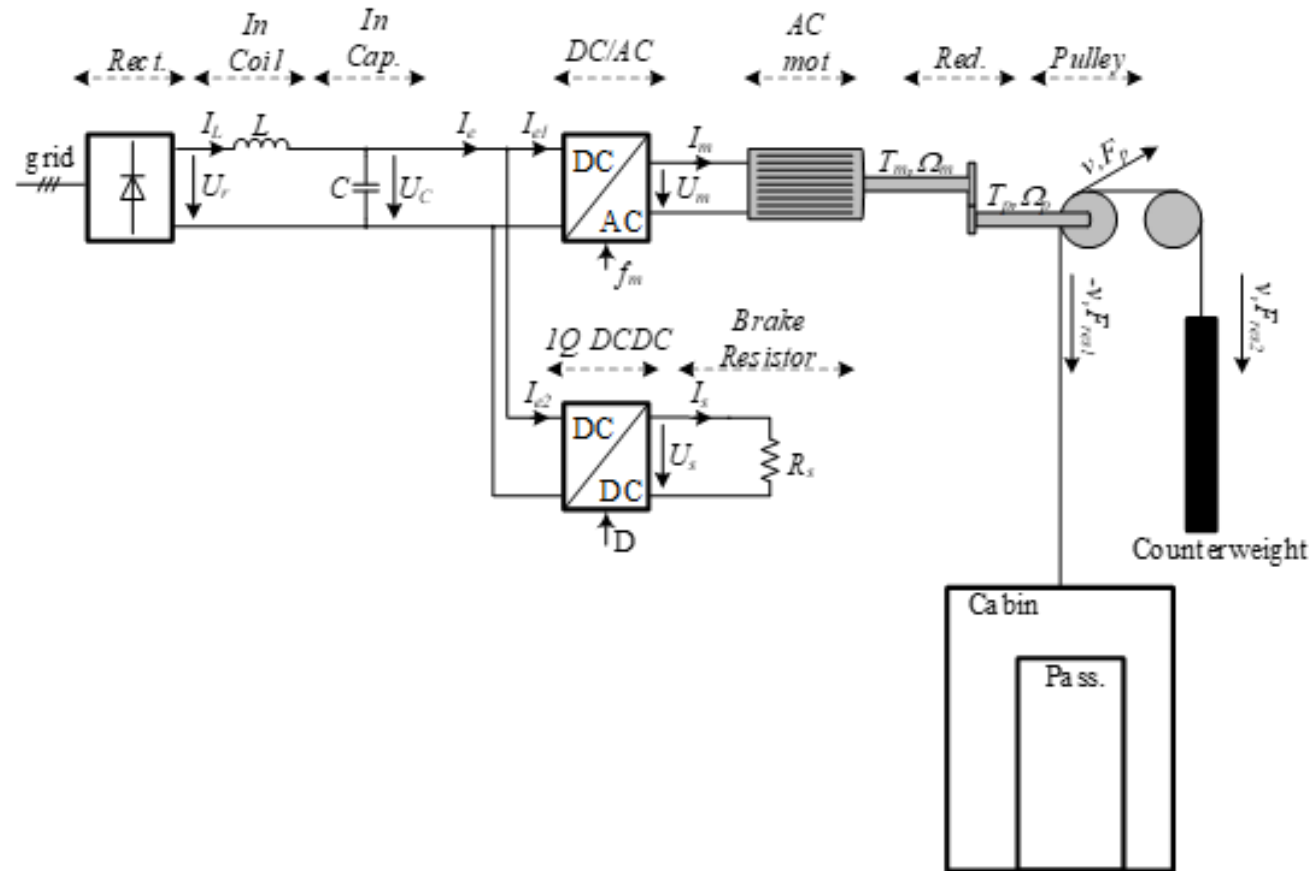


Technical Management

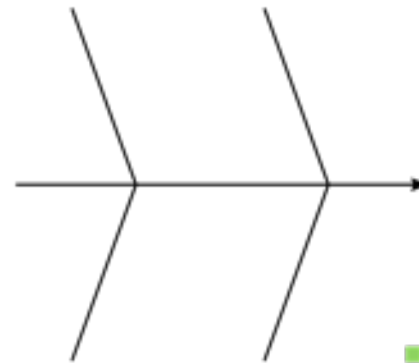
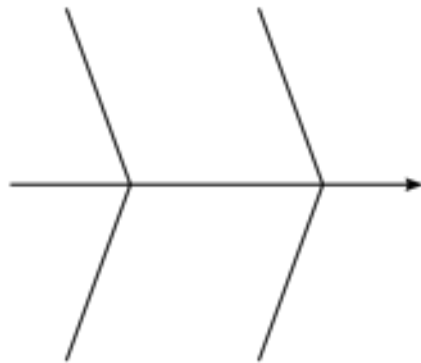
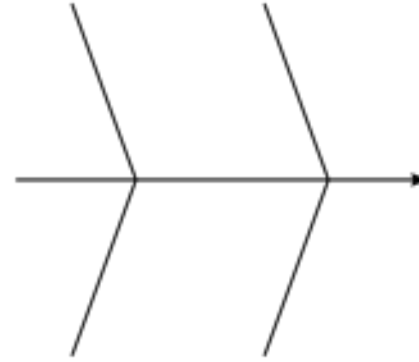
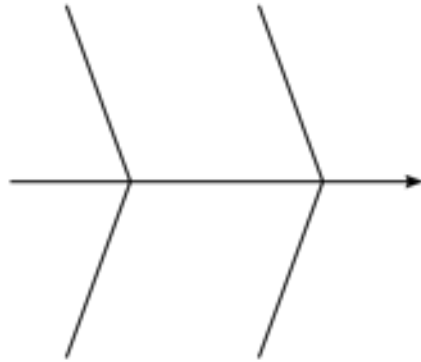


Example

Generic scheme of the system "Elevator"

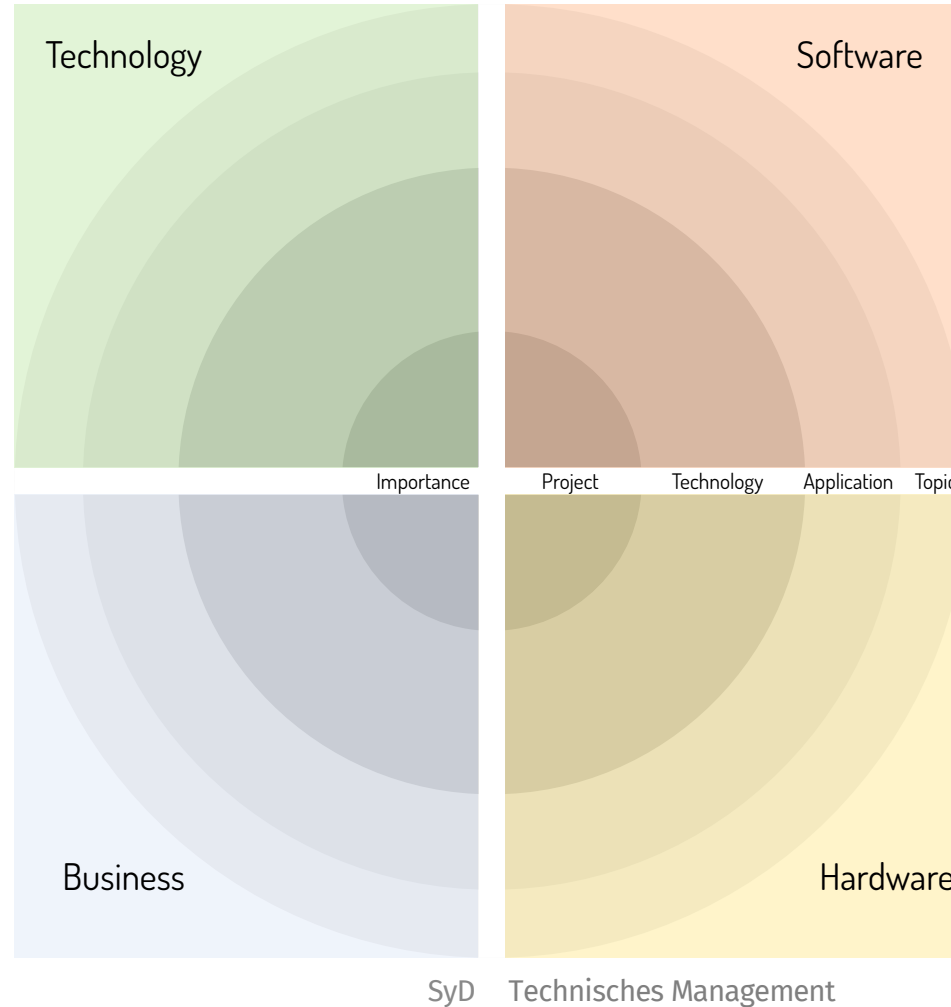


Cause and Effect Diagram



■ controllable factor
■ uncontrollable factor

Keyword Radar



UNCLASSIFIED

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Use the information as you see fit.

4/2/2008

SIMPLE SABOTAGE
FIELD MANUAL

Strategic Services
(Provisional)

Sabotage Field Manual

STRATEGIC SERVICES FIELD MANUAL No. 3

UNCLASSIFIED

Project Interference



(11) *General Interference with Organizations and Production*

(a) Organizations and Conferences

(1) Insist on doing everything through "channels." Never permit short-cuts to be taken in order to expedite decisions.

(2) Make "speeches." Talk as frequently as possible and at great length. Illustrate your "points" by long anecdotes and accounts of personal experiences. Never hesitate to make a few appropriate "patriotic" comments.

(3) When possible, refer all matters to committees, for "further study and consideration." Attempt to make the committees as large as possible — never less than five.

(4) Bring up irrelevant issues as frequently as possible.

(5) Haggle over precise wordings of communications, minutes, resolutions.

(6) Refer back to matters decided upon at the last meeting and attempt to re-open the question of the advisability of that decision.

(7) Advocate "caution." Be "reasonable" and urge your fellow-conferrees to be "reasonable" and avoid haste which might result in embarrassments or difficulties later on.

(8) Be worried about the propriety of any decision — raise the question of whether such action as is contemplated lies within the jurisdiction of the group or whether it might conflict with the policy of some higher echelon.

1. Open and direct communication
2. Be brief and precise
3. Direct line of communication
4. Wording is not important but understanding
5. Be fearless and bold